FEVEN ABEBE BEJIGA

Applied AI/ML Engineer | Responsible AI | Digital Health | Global Health and Development
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PROFESSIONAL SUMMARY

Candidate for the AI Fellow position in the Gates Foundation's Africa Regional Offices. Applied AI/ML practitioner with a BSc in Health Informatics (CGPA 3.86/4.00), an MSc in Artificial Intelligence in progress at Addis Ababa University, and 3+ years of digital health experience implementing EMR, DHIS2, and clinical decision-support workflows in Ethiopian public health facilities. Builds retrieval-augmented LLM, computer vision, and forecasting prototypes with attention to data quality, explainability, uncertainty, and human oversight. Trained healthcare staff, wrote technical documentation, and contributed to e-learning content for digital health leaders. First author of a federated learning paper accepted for presentation at PanAfriCon AI 2026.

TECHNICAL SKILLS

- **Applied AI and LLMs:** Retrieval-augmented generation (RAG), Gemini API, prompt engineering, FAISS, Sentence Transformers, semantic search, computer vision, NLP, time-series forecasting
- **Responsible AI:** Fairness metrics (coursework), explainability (SHAP, LIME), calibration, conformal prediction, out-of-distribution detection, human-in-the-loop design, data validation, EMR access control
- **ML and Data Science:** Python, PyTorch, TensorFlow/Keras, scikit-learn, LightGBM, CatBoost, pandas, NumPy, SQL, DuckDB, cross-validation, model evaluation
- **Software Engineering:** FastAPI, REST APIs, Docker, Git/GitHub, GitHub Actions (CI/CD), pytest unit and API testing, Streamlit
- **Digital Health:** DHIS2, Bahmni, EMR, WHO Digital Adaptation Kits, clinical decision support systems, indicator reporting, MySQL
- **Capacity Building:** Workshops and user training, technical documentation, e-learning content development, stakeholder communication, cross-functional collaboration

PROFESSIONAL EXPERIENCE

Big Data / Data Science Intern | Ethiopian Statistical Service (ESS) | October 2026 – Present

- **Data reconciliation:** Reconciling statistical figures across ESS publications and source documents using Python, DuckDB, and SQL to assess consistency between reported values.
- **Data provenance and validation:** Developing a provenance-focused workflow to extract, structure, validate, and compare statistical observations while maintaining traceability to original source tables.
- **SQL and quality assurance:** Implementing SQL-based transformations and validation checks to identify discrepancies, missing values, and inconsistencies in reported data.

Digital Health Developer | Center for Digital Health and Implementation (CDHi), University of Gondar | July 2023 – January 2026

- **Capacity building:** Delivered workshops and training on WHO DAK workflows and the hospital EMR to 100+ nurses, physicians, and other staff, with troubleshooting support; wrote user guidance and technical documentation for every system I customized.
- **Collaboration:** Worked with clinical, administrative, and technical teams at 3 healthcare facilities to implement and maintain Bahmni and DHIS2, translating facility workflows, user needs, and reporting requirements into system configuration.
- **Clinical decision support:** Co-implemented WHO Digital Adaptation Kit clinical workflows in the EMR at 3 facilities (University of Gondar Comprehensive Specialized Hospital, Han Health Center, Shimbit Health Center), configuring decision logic, validation rules, mandatory fields, skip logic, and automated calculations.
- **Data and reporting:** Authored an SQL EMR reporting module used at 3 facilities, generating aggregated indicators for 8+ service areas (ANC, FP, NCD, ambulance, inpatient, outpatient, PICT, TB); administered MySQL databases to safeguard data integrity.
- **DHIS2 configuration and access management:** Configured DHIS2 dashboards and reporting across 3 workstreams (Health and Demographic Surveillance System (HDSS) Tracker workflows, Informatics-Savvy Health Organization (ISHO) aggregate reporting, facility assessment data imported from Excel) with validation and program rules; managed EMR user access control (authentication, authorization, access rights).
- **Conversational support:** Contributed to integrating a chatbot into the ISHO system that gives users guidance on completing forms, supporting data-entry workflows.

Digital Health Intern / Fellow | Center for Digital Health and Implementation (CDHi), University of Gondar | November 2022 – June 2023

- **Implementation support:** Tested and documented digital health system implementations; provided user support, data validation, system configuration, and troubleshooting alongside healthcare professionals and technical teams.

PROFESSIONAL CONTRIBUTIONS

Contributor, Digital Health Leadership E-learning Course Development | International Institute for Primary Health Care – Ethiopia (IPHC) | November 2025 – February 2026

- Participated in in-person workshops and online collaborative activities to identify digital health leadership competency gaps and develop e-learning content for digital health leaders.
- Responsible for Module 4, Manage Digital Health Tools & Solutions, covering leading digital health initiatives (governance, stakeholder engagement, implementation roadmaps) and project management in digital health (methodologies, accountability frameworks, KPIs, risk management).

- Developed learning objectives, African digital health case studies, practical exercises, session summaries, and assessment questions for the module.

SELECTED AI/ML PROJECTS

Applied AI/ML work: 2024–present, spanning independent engineering, MSc coursework, research, and hackathon projects.

AgriMind: Evidence-Grounded LLM Decision Support for Crop Disease (early-stage prototype)
Python · TensorFlow/Keras · EfficientNet · Gemini API · Sentence Transformers · FAISS · Streamlit
- Built a RAG prototype combining maize and wheat disease classification (EfficientNet) with FAISS semantic retrieval and the Gemini API to give agricultural extension workers evidence-grounded information.
- Curated a 29-record knowledge base from 8 sources, organized by year, region, evidence type, and confidence; retrieval prioritizes evidence for the predicted disease and crop.
- Designed prompts and a response workflow that ground answers in evidence, communicate uncertainty, avoid unsupported pesticide recommendations, and refer users to experts when evidence is insufficient. Not yet field- or expert-validated.

CoffeeGuard AI: Coffee Leaf Disease Detection with Reliability Safeguards (team training project)
Python · PyTorch · EfficientNetV2-B0 · ONNX Runtime · FastAPI · Streamlit · Docker
- Contributed to training an EfficientNetV2-B0 model for four coffee-leaf classes and helped oversee data cleaning; the team cut a 12,000-file public dataset to 2,520 unique photos and used group-aware splits so no leaf crosses train, validation, and test.
- **Reliability and uncertainty:** Team model uses temperature scaling (ECE 0.097 to 0.014), conformal prediction, and a KNN out-of-distribution gate that rejected 180 of 182 non-coffee images; macro-F1 0.974 on 379 held-out photos. Accuracy fell to 63% on another dataset; not yet validated on farm photos.

DHIS2 Metadata Intelligence (prototype)
Python · FastAPI · React · TypeScript · SQLAlchemy · Sentence Transformers · Docker
- Created a read-only prototype that flags potentially duplicate DHIS2 metadata and shows the downstream effects of changing or deleting it, combining 4 similarity signals with dependency analysis.
- Designed a human-review workflow separating potential matches from confirmed decisions, with an evaluation framework to compare similarity approaches.

Ride Demand Forecasting: Qiyas Data Science & AI Hackathon (team of 6; among the top 3 teams)
Python · pandas · LightGBM · ARIMA · FastAPI · Next.js
- Co-developed hourly demand forecasting across 12 city zones on 10 months of synthetic ride-hailing data; corrected 28 data-quality issues, including a UTC-to-local-time mismatch that raised the rain–demand correlation from 0.02 to 0.27.
- Compared 10 approaches with rolling-origin validation; a tuned LightGBM model reached RMSE 8.09 on a held-out fortnight, 19% better than the seasonal-naive baseline. Synthetic data; experimental results.

House Price Prediction System (end-to-end project)
Python · scikit-learn · CatBoost · SHAP · FastAPI · Streamlit · Docker · GitHub Actions
- Built a pipeline on the Ames housing dataset: benchmarked 12 regression algorithms and tuned CatBoost with 5-fold grid search (R-squared 0.937 on a 292-home held-out set), with SHAP explainability.
- Served through FastAPI and Streamlit in Docker; wrote pytest unit and API tests, with GitHub Actions CI that runs the tests and publishes the Docker images.

RESEARCH AND ACADEMIC WORK

Lightweight Metaheuristic Optimization for Client Selection in Resource-Constrained Federated Learning Systems
Bejiga, F. A., Yassin, Z., and Gizachew, B. · First author; accepted for presentation at PanAfriCon AI 2026 · July 2026 – Present
- Implemented and evaluated 3 metaheuristics (Genetic Algorithm, Particle Swarm Optimization, Simulated Annealing) for client selection against a random baseline in PyTorch federated learning (FedProx, non-IID data) on FEMNIST, CIFAR-10, and CIFAR-100, with 5 random seeds and 100 communication rounds.

Fairness Prioritization Framework for Machine Learning in Finance and Healthcare
MSc Coursework Research Paper | Group Assignment | November 2025 – January 2026
- Examined algorithmic fairness in finance and healthcare ML, comparing fairness definitions, bias-detection metrics, and mitigation techniques across 16 research publications.
- Developed a context-sensitive fairness prioritization framework that considers equal treatment, disparate impact, regulatory compliance, and explainability in financial decision-making, alongside patient safety, ethical considerations, and equitable outcomes in healthcare.
- Analyzed trade-offs between predictive performance, fairness objectives, and interpretability when applying bias mitigation methods in high-impact decision-making.
  Scope: *Literature-based conceptual research; not a published study, formal fairness audit, or deployed system.*

EDUCATION

MSc in Artificial Intelligence, Addis Ababa University (AAiT) | November 2024 – Present
Proposed thesis: Selective Neuro-Symbolic Prediction with Proof-Based Acceptance under Concept and Rule Shift
BSc in Health Informatics, University of Gondar | CGPA 3.86/4.00 | 2022

PROFESSIONAL TRAINING

- Data Science and AI Engineering, Qiyas, Addis Ababa University | May – September 2026
- SAFEE KAIM, Kifiya AI Mastery Training Program, 10 Academy Women-Only Cohort | June – August 2026
- GCI World, Data Science, Matsuo-Iwasawa Laboratory, University of Tokyo | June – August 2026

Languages: Amharic (Native) | English (Fluent)